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MTH643 TODAY FINAL PAPER

SOLVED BY SANAULLAH KHAN

CONTACT NO:03066793776

Subject Enrolment

1. Online classes are available (Mathematics)
2. Solved graded activities
   - Assignment's
   - Quizzes
   - GDB's
3. Solved quiz files
4. Past papers
5. Any info about virtual university of Pakistan

Final project (MTH620)

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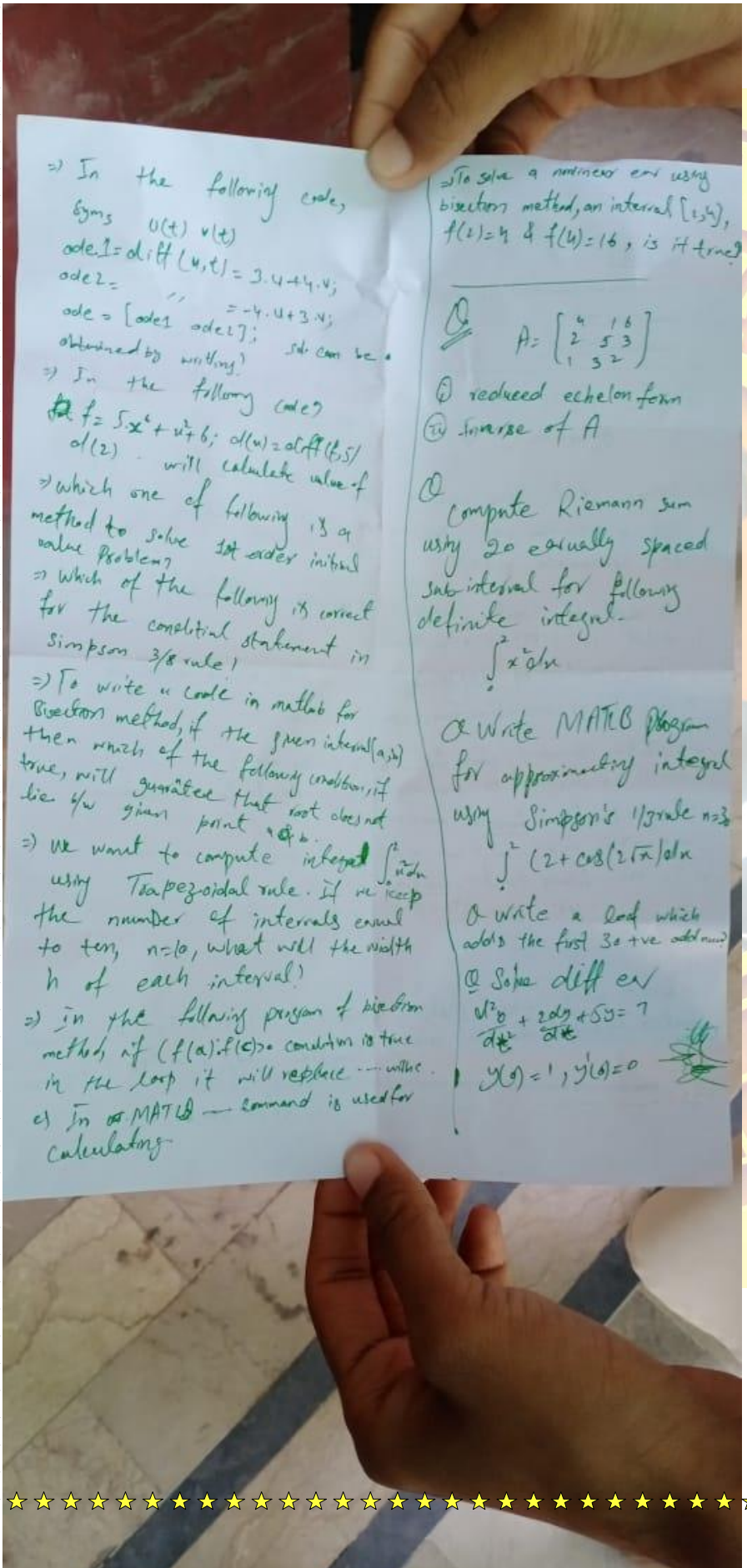
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For Mcq's See Youtube channel

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Question No 1: Matrix

Code:

```
clc
clear all
close all
A=[4 1 6;2 5 3;1 3 2];
% row reduce echolen form
D=rref(A)
%invers of A
E=inv(A)
```

Output:

D =

1	**0**	**0**
0	**1**	**0**
0	**0**	**1**

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E =

```
    0.1111    1.7778   -3.0000
   -0.1111    0.2222         0
    0.1111   -1.2222    2.0000
```

>>

Question No 2: Reimann sum

Code:

```
%Reimann sum
clc
close all
clear all
f=@(x)x^2;
a=0;
b=2;
n=20;
dx=(b-a)/n;
sum=0;
for i=1:n;
 c=a+i*dx;
 sum=sum+f(c);
end
Area=sum*dx
```

Output:

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Area =

2.8700

>>

Question No 3: Simpson's 1/3 Rule

Code:

```
%simpson1/3
clc
clear all
close all
f=@(x)(2+cos(2*sqrt(x)));
a=0;
b=2;
n=30;
h=(b-a)/n;
sum1=0;
sum2=0;
for k=1:n-1
x(k)=a+k*h;
if mod(k,2)==0
    sum1=sum1+f(x(k));
else
    sum2=sum2+f(x(k));
end
end
answer=h/3*(f(a)+f(b)+2*sum1+4*sum2)
```

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Output:

answer =

3.4600

>>

Question No 4: For Loop

Code:

```
clc
clear all
close all
%firt 30 od no 1 3 5 7 9 .... 59   so n=60
n=60;
sum=0;
for i=1:2:n
    sum=sum+i;
end
disp('Sum of first 30 +ve odd number ='),disp(sum)
```

Output:

Sum of first 30 +ve odd number =

900

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Question No 5: Differential equation

Code:

```
clc
clear all
close all
syms y(t)
ode=diff(y,t,2)+2*diff(y,t)+5*y==7;
dy=diff(y,t);
cond1=y(0)==1;
cond2=dy(0)==0;
cond=[cond1 cond2];
sol=dsolve(ode,cond)
```

Output:

sol =

7/5 - (sin(2*t)*exp(-t))/5 - (2*cos(2*t)*exp(-t))/5

>>

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12:00pm

Today Mth 643 27-02-2023

1) Write a program which takes a value from user and then using for loop find factorial.

2) Write a function name cube takes a number as a input if the number is even return cube of that number if odd then return square.

3) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 7$ and $y(0) = 1$ and $y'(0) = 0$ Solve the differential equation.

4) Solve the equation

$= 3e^{-t}$ by using trapezoidal rule interval [4 6] by using 10 sub interval.

5) ~~Find the value~~

Two matrix is given in linear form we can write that matrix in MATLAB and also write in reduce echelon form.

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Question No 1: Factorial

Code:

```
clc
clear all
close all
n=input('enter a number:');
i=n;
fact=1;
if i==0
fact=1
else if i<0
disp('the number you are entered is nagative')
    else
 while i>0
fact=fact*i;
i=i-1;
 end
disp('fact='),disp(fact)
    end
end
```

Output:

enter a number:7

fact=

5040

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Question No 2: function

Code:

```
function[cube square]=cube(n)
if rem(n,2)==1
square=n^2
else
cube=n^3
end
end
```

Output:

>> cube(4)

cube =

64

ans =

64

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Question No 3: Differential equation

Code:

Already don in above paper

Output:

Question No 4: Trapozidal rule

Code:

```
%trapozidl rule
clc
clear all
close all
f=@(t)(3*exp(-t));
a=4;
b=6;
n=10;
h=(b-a)/n;
sum=0;
for i=1:n-1
    t(i)=a+i*h;
    sum=sum+f(t(i));
end
answer=h/2*(f(a)+f(b)+2*sum)
```

Output:

answer =

0.0477

>>

Question No 5: Matrix

Code:

Already don in Above Paper

Output:

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